1.GET IP ADDRESS OF A DEVICE

```
<?xml version="1.0" encoding="UTF-8"?>-<manifest
package="com.arioninfotech.getipaddressexample"
xmlns:android="http://schemas.android.com/apk/res/android">

<uses-permission android:name="android.permission.ACCESS_WIFI_STATE"/>

-<application android:theme="@style/AppTheme" android:supportsRtl="true"
android:roundIcon="@mipmap/ic_launcher_round" android:label="@string/app_name"
android:icon="@mipmap/ic_launcher" android:allowBackup="true">

-<activity android:name=".MainActivity">

-<intent-filter>

<action android:name="android.intent.action.MAIN"/>

<category android:name="android.intent.category.LAUNCHER"/>

</intent-filter>

</activity>

</application>

</manifest>
```

MAIN ACTIVITY.JAVA

```
package com.arioninfotech.getipaddressexample;

import androidx.appcompat.app.AppCompatActivity;

import android.net.wifi.WifiManager;

import android.os.Bundle;

import android.text.format.Formatter;

import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    TextView lblIPAddress;

    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

        lblIPAddress = findViewById(R.id.lblIPAddress);
```

```
        WifiManager wifiManager = (WifiManager)
getApplicationContext().getSystemService(WIFI_SERVICE);

        String ipAddress =
Formatter.formatIpAddress(wifiManager.getConnectionInfo().getIpAddress());

        lblIPAddress.setText("Your Device IP Address: " + ipAddress);



    }}
```

```
<?xml version="1.0" encoding="UTF-8"?>

-<androidx.constraintlayout.widget.ConstraintLayout tools:context=".MainActivity"

android:layout_height="match_parent"

android:layout_width="match_parent"

xmlns:tools="http://schemas.android.com/tools" xmlns:app="http://schemas.android.com/apk/res-
auto"

xmlns:android="http://schemas.android.com/apk/res/android">

<TextView android:layout_height="wrap_content"

android:layout_width="0dp"

app:layout_constraintTop_toTopOf="parent"

app:layout_constraintStart_toStartOf="parent"

app:layout_constraintEnd_toEndOf="parent"

 app:layout_constraintBottom_toBottomOf="parent"

android:textSize="18sp"

android:textColor="@color/colorAccent"

 android:text="TextView" android:layout_marginBottom="474dp"

android:layout_marginRight="20dp"

 android:layout_marginEnd="20dp"

 android:layout_marginTop="174dp"

android:layout_marginLeft="16dp"

 android:layout_marginStart="16dp"

android:id="@+id/lblIPAddress"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

2.ALERTBOX

```
package com.arioninfotech.alertboxexample;

import androidx.appcompat.app.AlertDialog;

import androidx.appcompat.app.AppCompatActivity;

import android.content.DialogInterface;

import android.os.Bundle;

import android.view.View;

import android.widget.Button;

public class MainActivity extends AppCompatActivity {

   Button btnAlert;

   AlertDialog.Builder alertDialogBuilder ;

   @Override

   protected void onCreate(Bundle savedInstanceState) {

      super.onCreate(savedInstanceState);

      setContentView(R.layout.activity_main);

      btnAlert=findViewById(R.id.btnAlert);

      alertDialogBuilder = new AlertDialog.Builder(MainActivity.this);

      btnAlert.setOnClickListener(new View.OnClickListener() {

         @Override

         public void onClick(View view) {

            // set title

            alertDialogBuilder.setTitle("Confirmation Alert");

            // set dialog message

            alertDialogBuilder

                  .setMessage("Are you sure want to exit ? ")

                  .setCancelable(false)

                  .setPositiveButton("OK",new DialogInterface.OnClickListener() {

                     public void onClick(DialogInterface dialog, int id) {

                        // if this button is clicked, close

                        // current activity
```

```
                MainActivity.this.finish();
            }
        })
        .setNegativeButton("Cancel",new DialogInterface.OnClickListener() {
            public void onClick(DialogInterface dialog,int id) {
                // if this button is clicked, just close
                // the dialog box and do nothing
                dialog.cancel();
            }
        });

    // create alert dialog
    AlertDialog alertDialog = alertDialogBuilder.create();

    // show it
    alertDialog.show();

  }
 }); }}
```

3.ACTIVITY PROGRAM

main activity.java

```
package com.arioninfotech.intentexample;
import androidx.appcompat.app.AppCompatActivity;
import android.content.Intent;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
public class MainActivity extends AppCompatActivity {
   Button btnNext;
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btnNext=findViewById(R.id.btnNext);
        btnNext.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                Intent intent=new Intent(getApplicationContext(),
                        SecondActivity.class);
                startActivity(intent);
            }
        });
    } }
```

second activity .java

```
package com.arioninfotech.intentexample;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
public class SecondActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);
    }
}
```

main.xml

```
<?xml version="1.0" encoding="UTF-8"?>
-<androidx.constraintlayout.widget.ConstraintLayout tools:context=".MainActivity"
android:layout_height="match_parent"
android:layout_width="match_parent" xmlns:tools="http://schemas.android.com/tools"
```

```
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:android="http://schemas.android.com/apk/res/android">


<Button android:layout_height="48dp"

android:layout_width="349dp"

app:layout_constraintTop_toTopOf="parent"

app:layout_constraintStart_toStartOf="parent"

app:layout_constraintEnd_toEndOf="parent"

app:layout_constraintBottom_toBottomOf="parent"

android:text="Next " android:id="@+id/btnNext"/>

</androidx.constraintlayout.widget.ConstraintLayout>

<?xml version="1.0" encoding="UTF-8"?>

-<androidx.constraintlayout.widget.ConstraintLayout tools:context=".SecondActivity"

android:layout_height="match_parent" android:layout_width="match_parent"

xmlns:tools="http://schemas.android.com/tools"

 xmlns:app="http://schemas.android.com/apk/res-auto"

xmlns:android="http://schemas.android.com/apk/res/android">

<TextView android:layout_height="61dp"

android:layout_width="321dp"

app:layout_constraintTop_toTopOf="parent"

app:layout_constraintStart_toStartOf="parent"

app:layout_constraintEnd_toEndOf="parent"

app:layout_constraintBottom_toBottomOf="parent"

android:textSize="36sp" android:textColor="#FC1717"

android:text="Second Activity"

android:layout_marginRight="40dp"

android:layout_marginEnd="40dp"

android:layout_marginLeft="50dp"

android:layout_marginStart="50dp"

android:id="@+id/textView2"/>
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

4.SPINNER VIEW

```
<?xml version="1.0" encoding="UTF-8"?>
-<androidx.constraintlayout.widget.ConstraintLayout tools:context=".SecondActivity"
android:layout_height="match_parent"
android:layout_width="match_parent"
xmlns:tools="http://schemas.android.com/tools"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:android="http://schemas.android.com/apk/res/android">
<TextView android:layout_height="61dp"
android:layout_width="321dp"
app:layout_constraintTop_toTopOf="parent"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintEnd_toEndOf="parent"
app:layout_constraintBottom_toBottomOf="parent"
android:textSize="36sp" android:textColor="#FC1717" a
ndroid:text="Second Activity"
android:layout_marginRight="40dp"
 android:layout_marginEnd="40dp"
android:layout_marginLeft="50dp"
android:layout_marginStart="50dp"
android:id="@+id/textView2"/>
</androidx.constraintlayout.widget.ConstraintLayout>
```

ACTIVITY.JAVA

```
package com.example.spinnerexample;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
```

```
import android.view.View;

import android.widget.AdapterView;

import android.widget.ArrayAdapter;

import android.widget.Spinner;

import android.widget.Toast;


public class MainActivity extends AppCompatActivity {

   Spinner cmbSpinner;

   String[] languageNames={"Android","Java","Python","Angular"};

   @Override

   protected void onCreate(Bundle savedInstanceState) {

      super.onCreate(savedInstanceState);

      setContentView(R.layout.activity_main);

      cmbSpinner=findViewById(R.id.cmbSpinner);

      ArrayAdapter aa=new ArrayAdapter(this,
android.R.layout.simple_spinner_item,languageNames);

      aa.setDropDownViewResource(android.R.layout.simple_spinner_dropdown_item);

      cmbSpinner.setAdapter(aa);

      cmbSpinner.setOnItemSelectedListener(new AdapterView.OnItemSelectedListener() {

         @Override

         public void onItemSelected(AdapterView<?> adapterView, View view, int i, long l) {

            Toast.makeText(getApplicationContext(),languageNames[i],Toast.LENGTH_LONG ).show();

         }

         @Override

         public void onNothingSelected(AdapterView<?> adapterView) {


         }

      });


      }   }
```

5.OPTION MENU

```
package com.arioninfotech.optionmenuexample;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;

import android.view.Menu;

import android.view.MenuItem;

import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

    }

    @Override

    public boolean onCreateOptionsMenu(Menu menu) {

        // Inflate the menu; this adds items to the action bar if it is present.

        getMenuInflater().inflate(R.menu.menu_file, menu);

        return true;

    }

    @Override

    public boolean onOptionsItemSelected(MenuItem item) {

        int id = item.getItemId();

        switch (id){

            case R.id.item1:

                Toast.makeText(getApplicationContext(),"Item 1 Selected",Toast.LENGTH_LONG).show();

                return true;

            case R.id.item2:

                Toast.makeText(getApplicationContext(),"Item 2 Selected",Toast.LENGTH_LONG).show();

                return true;

            case R.id.item3:
```

```
            Toast.makeText(getApplicationContext(),"Item 3 Selected",Toast.LENGTH_LONG).show();
            return true;
        default:
            return super.onOptionsItemSelected(item);
    }
  }
}
```

XML

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity"/>
```

MENU CODE

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:android="http://schemas.android.com/apk/res/android">

    <item
        android:id="@+id/item1"
        android:title="menuItem1" />
    <item
        android:id="@+id/item2"
        android:title="menuItem2" />
    <item
        android:id="@+id/item3"
        android:title="menuItem3" />
</menu>
```

6.LIST VIEW

```
package com.arioninfotech.listviewexample;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;

import android.widget.ArrayAdapter;

import android.widget.ListView;

public class MainActivity extends AppCompatActivity {

   String[] countries = {"India","Pakistan","Afganistan","Bangladesh","China"};

   ListView lstCountries;
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        lstCountries=findViewById(R.id.lstCountries);
        ArrayAdapter adapter = new ArrayAdapter<String>(this,
                android.R.layout.simple_list_item_1, countries);
        lstCountries.setAdapter(adapter);   }}
```

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <ListView
        android:id="@+id/lstCountries"
        android:layout_width="0dp"
        android:layout_height="674dp"
        android:layout_marginStart="31dp"
        android:layout_marginTop="24dp"
        android:layout_marginEnd="25dp"
        android:layout_marginBottom="33dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

7.RADIOBUTTON

```
package com.arioninfotech.radiobuttonexample;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.TextView;
public class MainActivity extends AppCompatActivity {
    RadioGroup radioGroup;
```

```
    RadioButton gender;
    TextView txtSelectedGender;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        radioGroup=(RadioGroup) findViewById(R.id.radioGroup);
        txtSelectedGender=(TextView)findViewById(R.id.txtSelectedGender);
        radioGroup.setOnCheckedChangeListener(new RadioGroup.OnCheckedChangeListener() {
            @Override
            public void onCheckedChanged(RadioGroup radioGroup, int checkedId) {
                gender  = radioGroup.findViewById(checkedId);
                txtSelectedGender.setText("Selected Gender :"+gender.getText().toString());


            }
        });
    }
}
```

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <RadioGroup
        android:id="@+id/radioGroup"
        android:layout_width="356dp"
        android:layout_height="96dp"
        android:layout_marginStart="23dp"
        android:layout_marginTop="100dp"
        android:layout_marginEnd="32dp"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent">

        <RadioButton
            android:id="@+id/radioMale"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
```

```
            android:text="male"
            tools:text="Male" />

        <RadioButton
            android:id="@+id/radioFemale"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:text="Female" />
    </RadioGroup>

    <EditText
        android:id="@+id/txtSelectedGender"
        android:layout_width="351dp"
        android:layout_height="103dp"
        android:layout_marginBottom="228dp"
        android:ems="10"
        android:inputType="textPersonName"
        android:text=""
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="@+id/radioGroup"
        app:layout_constraintHorizontal_bias="0.0"
        app:layout_constraintStart_toStartOf="@+id/radioGroup" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

8.IMAGE LOAD

JAVA

```
8. package com.example.imageexample;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

XML

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <ImageView
        android:id="@+id/imageView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="53dp"
        android:layout_marginEnd="74dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
```

```
        app:layout_constraintTop_toTopOf="parent"
        app:srcCompat="@drawable/flower" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

9.EDIT TEXT TO TEXT VIEW

```
package com.example.edittextdatadisplay;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {
EditText txtData;
TextView lblData;
Button btnView;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        txtData=findViewById(R.id.txtData);
        lblData=findViewById(R.id.lblData);
        btnView=findViewById(R.id.btnView);
        btnView.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                lblData.setText(txtData.getText());
            }
        });
    }
}
```

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/txtData"
        android:layout_width="302dp"
        android:layout_height="53dp"
        android:layout_marginStart="28dp"
        android:layout_marginTop="161dp"
        android:layout_marginEnd="81dp"
        android:ems="10"
        android:hint="Enter Data"
        android:inputType="textPersonName"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
```

```
        app:layout_constraintTop_toTopOf="parent" />

    <Button
        android:id="@+id/btnView"
        android:layout_width="290dp"
        android:layout_height="73dp"
        android:layout_marginTop="83dp"
        android:text="view"
        app:layout_constraintEnd_toEndOf="@+id/txtData"
        app:layout_constraintStart_toStartOf="@+id/txtData"
        app:layout_constraintTop_toBottomOf="@+id/txtData" />

    <TextView
        android:id="@+id/lblData"
        android:layout_width="284dp"
        android:layout_height="171dp"
        android:layout_marginTop="43dp"
        app:layout_constraintEnd_toEndOf="@+id/btnView"
        app:layout_constraintStart_toStartOf="@+id/btnView"
        app:layout_constraintTop_toBottomOf="@+id/btnView" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

10.BG IMAGE

```
package com.example.backgroundimageexample;

import androidx.appcompat.app.AppCompatActivity;
import androidx.constraintlayout.widget.ConstraintLayout;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {
ConstraintLayout constraintLayout;
Button btnChange;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        constraintLayout=findViewById(R.id.constraintLayout);
        btnChange=findViewById(R.id.btnChange);
        btnChange.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {

constraintLayout.setBackgroundResource(R.drawable.download);

            }
        });
    }
}
```

XML

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
```

```
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/constraintLayout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="@drawable/sun"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/btnChange"
        android:layout_width="278dp"
        android:layout_height="69dp"
        android:text="changeBackground"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.496"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintVertical_bias="0.835" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

11.ADDITION

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/txtnumber1"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="44dp"
        android:ems="10"
        android:inputType="textPersonName"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintHorizontal_bias="0.447"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <EditText
        android:id="@+id/txtnumber2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="116dp"
        android:ems="10"
        android:inputType="textPersonName"
        app:layout_constraintEnd_toEndOf="@+id/txtnumber1"
        app:layout_constraintHorizontal_bias="0.0"
        app:layout_constraintStart_toStartOf="@+id/txtnumber1"
        app:layout_constraintTop_toBottomOf="@+id/txtnumber1" />

    <EditText
        android:id="@+id/txtresult"
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:layout_marginTop="136dp"
        android:ems="10"
        android:inputType="textPersonName"
        app:layout_constraintEnd_toEndOf="@+id/txtnumber2"
        app:layout_constraintHorizontal_bias="1.0"
        app:layout_constraintStart_toStartOf="@+id/txtnumber2"
        app:layout_constraintTop_toBottomOf="@+id/txtnumber2" />

    <Button
        android:id="@+id/btnsum"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="68dp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="@+id/txtresult"
        app:layout_constraintStart_toStartOf="@+id/txtresult"
        app:layout_constraintTop_toBottomOf="@+id/txtresult" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

JAVA

```
package com.example.addition;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;


public class MainActivity extends AppCompatActivity {
EditText txtnumber1,txtnumber2,txtresult;
Button btnsum;
double a,b,c;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        txtnumber1=findViewById(R.id.txtnumber1);
        txtnumber2=findViewById(R.id.txtnumber2);
        txtresult=findViewById(R.id.txtresult);
        btnsum=findViewById(R.id.btnsum);
        btnsum=(Button) findViewById(R.id.btnsum);
        btnsum.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
               a=Double.parseDouble(txtnumber1.getText().toString());
               b=Double.parseDouble(txtnumber2.getText().toString());
               c=a+b;
               txtresult.setText(Double.toString(c));
            }
        });
    }
}
```

12.CHECK BOX

```java
package com.example.checkbox;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.CheckBox;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {
    CheckBox chkAndroid,chkJava,chkPython;
    Button btnShow;
    StringBuffer result;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        chkAndroid=findViewById(R.id.chkAndroid);
        chkJava=findViewById(R.id.chkJava);
        chkPython=findViewById(R.id.chkPython);
        btnShow=findViewById(R.id.btnShow);
        btnShow.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                result=new StringBuffer();
                if (chkAndroid.isChecked()) {
                    result.append(chkAndroid.getText() + " ");
                }
                if (chkJava.isChecked()){
                    result.append(chkJava.getText());
                }
                if (chkPython.isChecked()){
                    result.append(chkPython.getText());
                }
                Toast.makeText(getApplicationContext(),
                        "Checked Languages:"+result.toString(),
                        Toast.LENGTH_LONG).show();
            }
        });
    }
}
```

```androidx.constraintlayout.widget.ConstraintLayout
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <CheckBox
        android:id="@+id/chkAndroid"
        android:layout_width="330dp"
        android:layout_height="72dp"
        android:layout_marginStart="27dp"
        android:layout_marginTop="123dp"
        android:layout_marginEnd="54dp"
```

```
        android:text="Android"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <CheckBox
        android:id="@+id/chkJava"
        android:layout_width="306dp"
        android:layout_height="74dp"
        android:layout_marginTop="62dp"
        android:text="Java"
        app:layout_constraintEnd_toEndOf="@+id/chkAndroid"
        app:layout_constraintStart_toStartOf="@+id/chkAndroid"
        app:layout_constraintTop_toBottomOf="@+id/chkAndroid" />

    <CheckBox
        android:id="@+id/chkPython"
        android:layout_width="298dp"
        android:layout_height="57dp"
        android:layout_marginTop="54dp"
        android:text="Python"
        app:layout_constraintEnd_toEndOf="@+id/chkJava"
        app:layout_constraintStart_toStartOf="@+id/chkJava"
        app:layout_constraintTop_toBottomOf="@+id/chkJava" />

    <Button
        android:id="@+id/btnShow"
        android:layout_width="219dp"
        android:layout_height="67dp"
        android:layout_marginTop="26dp"
        android:text="SHOW"
        app:layout_constraintEnd_toEndOf="@+id/chkPython"
        app:layout_constraintStart_toStartOf="@+id/chkPython"
        app:layout_constraintTop_toBottomOf="@+id/chkPython" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

13. **Write a program to display multiplication table of a given number**

```
package com.arioninfotech.multiplicationtableexample;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity {
    EditText txtNumber;
    TextView lblResult;
    Button btnShow;
    int number=0,answer=0;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        txtNumber=findViewById(R.id.txtNumber);
```

```
        lblResult=findViewById(R.id.lblResult);
        btnShow=findViewById(R.id.btnShow);

        btnShow.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {

                if(txtNumber.getText().toString().isEmpty()){
                    Toast.makeText(getApplicationContext(),"Please
enter a number",Toast.LENGTH_LONG).show();

                }else{

                    StringBuffer buffer = new StringBuffer();


number=Integer.parseInt(txtNumber.getText().toString().trim());

                    // build the logic for table
                    for (int i = 1; i <= 10; i++) {
                        answer = (i * number);
                        buffer.append(number + " X " + i
                                + " = " + answer + "\n\n");
                    }

                    // set the buffer textview
                    lblResult.setText(buffer);
                }




            }
        });

    }
}
```

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/txtNumber"
        android:layout_width="244dp"
        android:layout_height="61dp"
        android:layout_marginStart="16dp"
        android:layout_marginTop="77dp"
        android:ems="10"
        android:hint="number"
        android:inputType="textPersonName"
        app:layout_constraintStart_toStartOf="parent"
```

```
        app:layout_constraintTop_toTopOf="parent" />

    <Button
        android:id="@+id/btnShow"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginTop="85dp"
        android:layout_marginEnd="31dp"
        android:text="Show"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <TextView
        android:id="@+id/txtResult"
        android:layout_width="392dp"
        android:layout_height="358dp"
        android:layout_marginTop="64dp"
        app:layout_constraintEnd_toEndOf="@+id/btnShow"
        app:layout_constraintHorizontal_bias="0.439"
        app:layout_constraintStart_toStartOf="@+id/txtNumber"
        app:layout_constraintTop_toBottomOf="@+id/txtNumber" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

14. **Write a program to Change the Background colour of the Activity**

```
package com.example.backgroundcolorexample;

import androidx.appcompat.app.AppCompatActivity;
import androidx.constraintlayout.widget.ConstraintLayout;

import android.graphics.Color;
import android.os.Bundle;
import android.view.View;
import android.view.View.OnClickListener;
import android.widget.Button;

public class MainActivity extends AppCompatActivity {
ConstraintLayout constraintLayout;
Button btnChange;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        constraintLayout=findViewById(R.id.constraintLayout);
        btnChange=findViewById(R.id.btnChange);
        btnChange.setOnClickListener(new OnClickListener() {
            @Override
            public void onClick(View view) {
                constraintLayout.setBackgroundColor(Color.RED);
            }
        });
    }
}
```

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
```

```
    android:id="@+id/constraintLayout"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/btnChange"
        android:layout_width="259dp"
        android:layout_height="64dp"
        android:text="Change BackgroundColor"
        tools:layout_editor_absoluteX="80dp"
        tools:layout_editor_absoluteY="511dp" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

15.TOAST

```
package com.arioninfotech.toastexample;

import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.Toast;

public class MainActivity extends AppCompatActivity {

    Button btnToast;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnToast=(Button)findViewById(R.id.button);
        btnToast.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {

                Toast.makeText(getApplicationContext(),
                        "Arion Infotech , KINFRA IT PARK",
                        Toast.LENGTH_LONG).show();

            }
        });
    }
}
```

XML

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
-<androidx.constraintlayout.widget.ConstraintLayout tools:context=".MainActivity"
android:layout_height="match_parent" android:layout_width="match_parent"
xmlns:tools="http://schemas.android.com/tools" xmlns:app="http://schemas.android.com/apk/res-
auto" xmlns:android="http://schemas.android.com/apk/res/android">


<Button android:layout_height="58dp" android:layout_width="347dp"
app:layout_constraintTop_toTopOf="parent" app:layout_constraintStart_toStartOf="parent"
app:layout_constraintEnd_toEndOf="parent" app:layout_constraintBottom_toBottomOf="parent"
android:text="Show Toast" android:layout_marginBottom="391dp"
android:layout_marginRight="32dp" android:layout_marginEnd="32dp"
android:layout_marginTop="282dp" android:layout_marginLeft="32dp"
android:layout_marginStart="32dp" android:id="@+id/button"/>


</androidx.constraintlayout.widget.ConstraintLayout>
```